P04: Makers Makin' It, Act II -- The Seequel

Design Doc by RustyRoosters, pd. 5

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PROJECT NAME: Fitness Genie
DESCRIPTION: Our website is a meal calorie calculator. Users can sign up/login, track a meal's calories through a db of known food calories, save them to their account, and view them later on a personal profile page. Users can look through premade graphs to determine how their meals are different

TARGET SHIP DATE: 4/20/2026

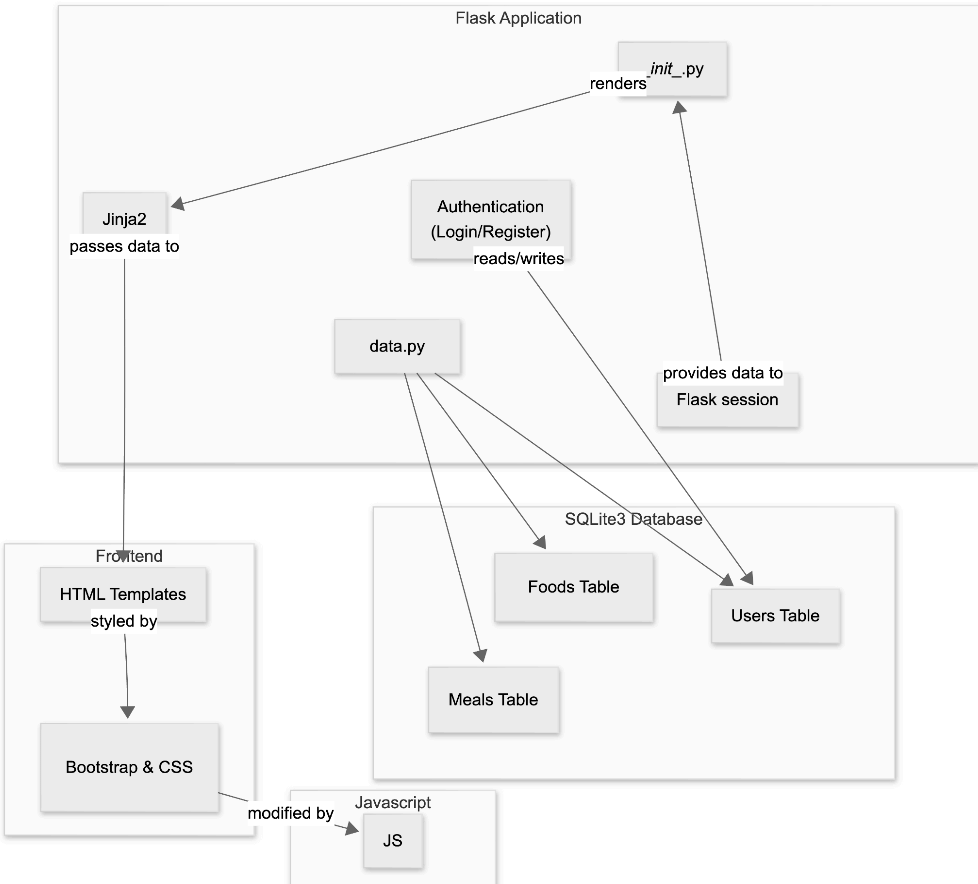
Program Components:

1) Flask App (Python)
    a) **__init__.py** (main app + routes)
        i) Creates Flask app, config, and session handling
        ii) Renders templates and connects frontend to backend logic
        iii) Routes:
            (1) **/register** adds a new user to the **users** table (**data.py**). Checks if the username is unique, hashes the password, stores it in session, then redirects to **/dashboard**.
            (2) **/login** checks users (**data.py**), verifies password hash, stores session, redirects to **/dashboard**.
            (3) **/logout** clears session, redirects to **/login**.
            (4) **/home** displays data on the nutrition of different foods as well as general calorie intake of a sample of 2000 people
            (5) **/calculate** allows users to select food that they eat in a day and it will calculate the calorie intake and compare it with the rest of the general data from the nutrition intake dataset by redirecting the to **/visualize**
            (6) **/visualize** Displays interactive graphs using **D3.js** based on:
                - User's calorie intake compared with general nutrition dataset
                - Macronutrient breakdown (carbs, protein, fat)
                  Users can filter by:
                - Nutrient type, etc.
            (7) **/profile/** uses **data.py** to load a creator's saved meals, calories, and nutrition stats
    b) data.py
        i) Connects to SQLite3 database and creates/maintains tables:
            (1) **users:** stores user account information.
            (2) **foods:** food name, calories, nutrients
            (3) **meals:** list of foods + user?
2) Database (SQLite3) (stored in data.db)
    a) **users** table stores all usernames and password hashes for authentication.
    b) **foods** nutritional data
    c) **meals** user meal logs
3) Frontend Framework
    a) **Bootstrap**: used for navbar, forms (login/registration), buttons, layout/grid, and cards for food items to keep styling consistent and responsive.
4) JavaScript

a) **calculate.js**: manages food selection and calorie calculations
b) **visualize.js**: renders D3.js graphs and handles data filtering
c) **home.js**: displays overview data and basic visualizations
d) **profile.js**: loads and displays user-specific nutrition statistics

5) RESTful APIs
    a) None
6) External CSS (if needed)

Component Map:



Database Organization

USERS			
TEXT	name	PK	Unique
TEXT	password		For authentication
DATE	created_at		

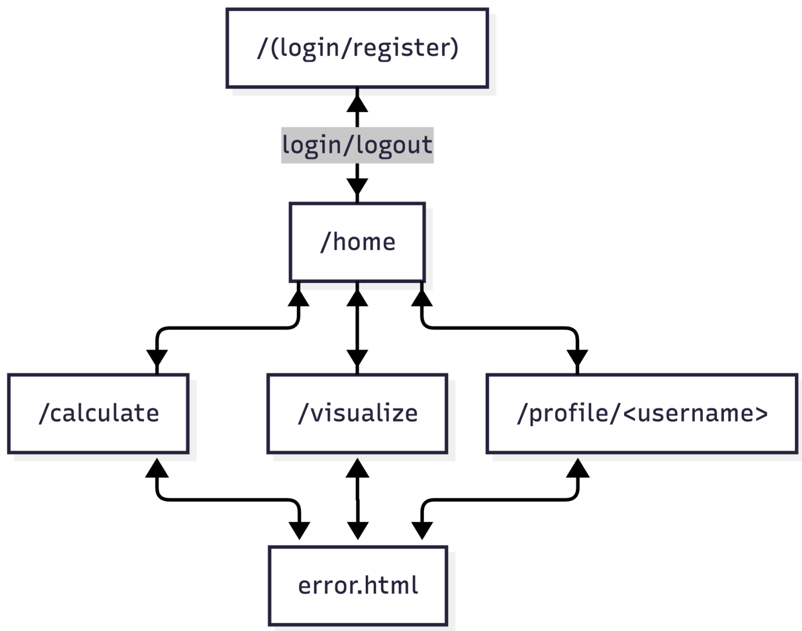
Foods			
INTEGER	id	PK	Auto-increment
TEXT	name		

INTEGER	calories		
FLOAT	protein		(grams)
FLOAT	carbs		(grams)
FLOAT	fat		(grams)

Meals			
INTEGER	id	PK	Auto-increment
TEXT	username	FK	USERS(username)
INTEGER	food_id	FK	FOODS(id)
FLOAT	quantity		

Site Map:

7) Templates (HTML)
   a) **/login** (username + password. Error message if login fails).
   b) **/register** (username + password. Error message if username is taken).
   c) **(home.html) /home** (Displays: Overview of nutrition data, basic D3 visualizations, Info about healthy calorie ranges)
   d) (**calculate.html**) **/calculate** Allows users to:
      - Select foods from database
      - Input quantity
      - Calculate total daily calorie intake
      - Compare result with general dataset
   e) (**visualize.html**) **/visualize** Interactive D3 graphs:
      - Line chart (calories over time or vs. average)
      - Bar chart (macronutrient breakdown)
      - Comparison chart (user vs dataset)
   f) (**profile.html**) **/profile/<username>** (shows username, nutritional stats).
   g) **error.html** (invalid id's or permission errors).



Visualization Library (D3 JS)

1. Line graphs (calorie intake over time)
2. Bar charts (macronutrients of meal)
3. Difference charts (user vs dataset averages)

APIs

(None needed)

Datasets

1. https://www.kaggle.com/datasets/abdussamad123/user-daily-nutritional-intake
    - Contains daily nutritional intake across a large sample of people which we will use to compare to the user's stats
2. https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-dataset/data
    - Contains nutritional values for specific food items to calculate the nutrients in the user's meals

Front End Framework

Bootstrap

Tasks & Assignments

TASK BREAKDOWN

Task	**Devo(s)**
Login & Register	James Sun
Editor UI & Javascript	Matthew Ciu
DB Organization	Yuhang Pan
D3JS Visualizations	Thomas Mackey